

The Dataset

This is our historical data. We have 10 students whose outcomes (Pass/Fail) are already known.

Student	Attendance(X)	Quiz(Y)	Class(Label)
S1	95	92	Pass
S2	55	45	Fail
S3	88	85	Pass
S4	60	50	Fail
S5	92	88	Pass
S6	45	40	Fail
S7	98	95	Pass
S8	50	55	Fail
S9	85	80	Pass
S10	65	60	Fail

New Entry:

Student	Attendance(X)	Quiz(Y)
S_New	80	75

Euclidean distance formula:

$$\sqrt{(x_2 - x_1)^2 + (Y_2 - Y_1)^2}$$

- X_2 = New entry's attendance (80).
- X_1 = Existing entry's attendance.
- Y_2 = New entry's quiz (75).
- Y_1 = Existing entry's quiz.

Distance Calculation:

For the first row, d1:

$$\begin{aligned}d1 &= \sqrt{(80 - 95)^2 + (75 - 92)^2} \\&= \sqrt{(-15)^2 + (-17)^2} \\&= \sqrt{225 + 289} \\&= \sqrt{514} \\&= 22.67\end{aligned}$$

For the second row, d2:

$$\begin{aligned}d2 &= \sqrt{(80 - 55)^2 + (75 - 45)^2} \\&= \sqrt{25^2 + 30^2} \\&= \sqrt{625 + 900} \\&= \sqrt{1525} \\&= 39.05\end{aligned}$$

For the third row, d3:

$$\begin{aligned}d3 &= \sqrt{(80 - 88)^2 + (75 - 85)^2} \\&= \sqrt{(-8)^2 + (-10)^2} \\&= \sqrt{64 + 100} \\&= \sqrt{164} \\&= 12.81\end{aligned}$$

For the fourth row, d4:

$$\begin{aligned}d4 &= \sqrt{(80 - 60)^2 + (75 - 50)^2} \\&= \sqrt{20^2 + 25^2} \\&= \sqrt{400 + 625} \\&= \sqrt{1025} \\&= 32.02\end{aligned}$$

For the fifth row, d5:

$$\begin{aligned}d5 &= \sqrt{(80 - 92)^2 + (75 - 88)^2} \\&= \sqrt{(-12)^2 + (-13)^2} \\&= \sqrt{144 + 169} \\&= \sqrt{313} \\&= 17.69\end{aligned}$$

For the sixth row, d6:

$$\begin{aligned}d6 &= \sqrt{(80 - 45)^2 + (75 - 40)^2} \\&= \sqrt{35^2 + 35^2} \\&= \sqrt{1225 + 1225} \\&= \sqrt{2450} \\&= 49.50\end{aligned}$$

For the seventh row, d7:

$$\begin{aligned}d7 &= \sqrt{(80 - 98)^2 + (75 - 95)^2} \\&= \sqrt{(-18)^2 + (-20)^2} \\&= \sqrt{324 + 400} \\&= \sqrt{724} \\&= 26.91\end{aligned}$$

For the eighth row, d8:

$$\begin{aligned}d8 &= \sqrt{(80 - 50)^2 + (75 - 55)^2} \\&= \sqrt{30^2 + 20^2} \\&= \sqrt{900 + 400} \\&= \sqrt{1300} \\&= 36.06\end{aligned}$$

For the ninth row, d9:

$$\begin{aligned}d9 &= \sqrt{(80 - 85)^2 + (75 - 80)^2} \\&= \sqrt{(-5)^2 + (-5)^2} \\&= \sqrt{25 + 25} \\&= \sqrt{50} \\&= 7.07\end{aligned}$$

For the tenth row, d10:

$$\begin{aligned}d10 &= \sqrt{(80 - 65)^2 + (75 - 60)^2} \\&= \sqrt{15^2 + 15^2} \\&= \sqrt{225 + 225} \\&= \sqrt{450} \\&= 21.21\end{aligned}$$

Sort in Ascending Order

Student	Distance	Class (Label)
S9	7.07	Pass
S3	12.81	Pass
S5	17.69	Pass
S10	21.21	Fail
S1	22.67	Pass
S7	26.91	Pass
S4	32.02	Fail
S8	36.06	Fail
S2	39.05	Fail
S6	49.50	Fail

Choose k?

If $k=5$ the algorithm looks at the three students with the shortest distance to your new entry. Based on your sorted data, these are:

Student	Distance	Class (Label)
S9	7.07	Pass
S3	12.81	Pass
S5	17.69	Pass
S10	21.21	Fail
S1	22.67	Pass

The Majority Vote

To classify the new point, we take a "majority vote" among these three neighbors:

- **Pass votes:** 3 (S9, S3, S5, S1)
- **Fail votes:** 1 (S10)

Because the majority of the five nearest neighbors are labeled "Pass" the algorithm classifies your new entry as **Pass**.